

VisionMetrics ESP32

VisionMetrics ESP32 is a lightweight visual testing system built for ESP32 devices with TFT displays. It presents controlled visual stimuli and records user responses to measure basic visual performance such as acuity, contrast sensitivity, and reaction time.

This is not a medical diagnostic tool. It is an experimental platform for structured visual testing and self-tracking under consistent conditions.

Purpose:

- Run repeatable visual perception tests
- Measure response accuracy and reaction time
- Track changes in visual performance over time
- Explore human visual processing metrics using embedded hardware

Core Test Modes:

1. Visual Acuity Test: orientation-based optotypes at varying sizes
2. Contrast Sensitivity Test: fixed-size targets at different brightness levels
3. Reaction Time Test: measures latency between stimulus and response

System Characteristics:

- ESP32 microcontroller
- SPI TFT display (ILI9341 / ST7789)
- Button-based input
- Lightweight real-time execution

Intended Use:

- Personal visual performance tracking
- Human-computer interaction experimentation
- Educational demonstration of perception measurement systems
- Low-cost vision testing prototyping

Limitations:

- Not medically calibrated
- Requires consistent viewing distance and lighting
- Provides relative, not clinical measurements

Concept Summary:

A controlled stimulus-response system where visual inputs are standardized and user responses are measured quantitatively over time.